

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10 - ELECTROSTATICS (3.4) - FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving Question: Where do electric charges come from? Why does a rubbed balloon attract or repel other objects? Write your own Final Explanation of the mysterious balloon. Use evidence from your lessons: the balloon-and-hair data (20 strokes, hair rising about 5 cm, balloon sticking to the wall for about 3 minutes, paper pieces jumping at about 2 cm), the rods and strips, the triboelectric order, the conduction test, the electroscope readings, and the humidity data from Nairobi and Mombasa. Answer each section in full sentences. Use the words electric charge, electron transfer, electrostatic force, conductor, insulator, induction and earthing correctly. Draw labelled particle-level diagrams where asked. Say which lesson or observation each piece of evidence comes from. Remember that only electrons move in solids.

Part 1: Where do electric charges come from?

Describe the balloon-and-hair phenomenon in Lesson 1 in your own words. Then explain where the electric charges on the balloon and hair come from. In your answer: (a) describe what atoms contain and why a normal atom is neutral, using the charge values you met; (b) explain what happens to electrons and protons when the balloon is rubbed on dry hair; (c) use the triboelectric order to say why the balloon becomes negative and the hair positive; (d) explain why we say charge is conserved and not created, using the number of electrons moved and the number of atoms in a balloon.

In Lesson 1, a latex balloon rubbed on dry hair for 20 strokes made the hair rise and follow it up to about 5 cm. The balloon then stuck to the wall for about 3 minutes, and 1 cm x 1 cm paper pieces jumped up to it from about 2 cm away. Two rubbed balloons on threads pushed apart, and an unrubbed balloon did nothing.

(a) Atoms have positive protons ($+1.6 \times 10^{-19}$ C) and neutral neutrons (0 C) in the nucleus, with negative electrons (-1.6×10^{-19} C) around it. A neutral atom has equal numbers of protons and electrons, so its total charge is 0 C.

(b) Friction transfers electrons from one material to the other. Electrons move from the hair to the balloon. Protons stay in the nucleus and do not move.

(c) In the triboelectric order (dry human hair (+), glass, perspex, wool/fur, silk, polythene, ebonite, latex balloon (-)), hair is higher than latex, so latex holds electrons more strongly. The balloon gains extra electrons and becomes negative. The hair has a deficit of electrons and becomes positive.

(d) The balloon gains about 1 nC (10^{-9} C), which is about 6×10^9 electrons. The hair loses the same number. The balloon has about 10^{23} atoms, so only a tiny fraction of electrons moves. Total charge stays the same, so charge is transferred, not created.

Part 2: Why does the balloon attract or repel? Hair, balloon and the rule of forces

Explain why (a) the two rubbed

The rods and strips gave the evidence for the rule. Two polythene strips rubbed with wool/fur repel, two perspex rods rubbed with

balloons on threads pushed apart, and (b) the hair strands rise and are pulled toward the balloon. Use the results from the rods and strips (polythene strip and perspex rod) as evidence for the rule you use. Then explain why the balloon keeps its charge for about 3 minutes on a dry day rather than losing it at once, using what you found in the conduction test with the 3 V cell and bulb. Name the type of force involved.	silk repel, and a polythene strip and a perspex rod attract. So like charges repel and unlike charges attract. This is the electrostatic force. (a) Each rubbed balloon gained extra electrons from the hair, so both are negative. Like charges repel, so they push apart. (b) The hair strands lost electrons, so every strand is positive. Like charges repel, so the strands push away from each other and rise. The positive hair and the negative balloon have unlike charges, so they attract and the hair follows the balloon up to about 5 cm. In the conduction test, the bulb lit with copper wire, aluminium foil, a steel spoon and graphite, which are conductors with free (delocalised) electrons. It did not light with a plastic ruler, dry wood, rubber, a glass rod or dry cotton cloth, which are insulators with tightly bound electrons. Latex is an insulator, so the extra electrons cannot move away. The balloon stays charged and sticks for about 3 minutes. A metal spoon is a conductor, so its charge would spread away and not stay.
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Part 3: Why do neutral paper and the wall get attracted? Contact, induction and earthing

The paper pieces and the wall are neutral, yet the balloon attracts them. (a) Using a particle-level diagram and words, explain why neutral paper and wall are attracted but never repelled. (b) Explain why some paper pieces fall off after touching the balloon, using what you saw with the electroscope and the polythene strip touching the cap. (c) Compare charging by contact with charging by induction with earthing: which sign does each give, and what does earthing do? Use the electroscope observations as evidence.	(a) The negative balloon repels electrons in the neutral paper and wall, so electrons shift slightly away. The near surface becomes positive and the far side negative, and this is induction. Diagram: balloon (-) on the left, then the wall or paper with + on the near side and - on the far side. The nearer positive charge is attracted more strongly than the farther negative charge is repelled. So there is a net attraction, and a neutral object is attracted but never repelled. This is why the paper jumps and the balloon sticks to the wall. (b) When a negative polythene strip touched the cap of a neutral electroscope, the leaf stayed at about 30 degrees after the strip was removed. Free electrons passed from the strip to the cap, stem and leaf, so they became negative, the same sign as the strip, and repelled each other. In the same way, a paper piece that touches the negative balloon gains electrons, becomes negative and is repelled, so it falls off. Touching the cap with a finger returned the leaf to 0 degrees. (c) Contact gives the SAME sign as the charged object, because electrons are transferred by touching. Induction with earthing gives the OPPOSITE sign. With the strip near but not touching, the leaf rose to about 30 degrees. Touching the cap with a finger (earthing) let electrons flow to the Earth, and the leaf fell to 0 degrees. After the finger and then the strip were removed, the leaf rose to about 30 degrees and stayed, showing a charge of opposite sign to the strip. In all cases only electrons move and charge is transferred, not created.
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Part 4: Charge on conductors, sparks, humidity and detecting charge

(a) Using the proof-plane and electroscope readings (hollow metal can; pear-shaped conductor), explain where charge sits on a conductor.	(a) On the hollow can, the proof plane on the outside gave about 40 degrees and on the inside 0 degrees. On the pear-shaped conductor the pointed end gave about 45 degrees and the flat end about 20 degrees. Free electrons on a conductor repel each other, so they move to the outer surface and crowd at points. At a point the strong field ionises the air and charge leaks away. This is point discharge. The jersey shock is a small spark when built-up charge discharges, and a faint crackle can be heard in the dark. A lightning conductor is a pointed metal rod on top of the building, joined by thick copper strip to a buried earth plate. It lets charge
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Then explain how this accounts for the small shock felt on a dry Nairobi day (for example, Wanjiru's shock after combing her hair) and for lightning conductors in Kericho and Kisumu. (b) Explain why Amina's balloon hardly works in Mombasa but works well in dry Nairobi, using the humidity figures. (c) Explain how a gold-leaf electroscope shows that the balloon is charged, how you can tell the sign of a charge, and what a bigger leaf angle tells you.	leak away at the point and lets electrons flow safely to the ground, protecting buildings in Kericho and Kisumu. (b) At about 35% humidity in dry Nairobi the balloon sticks for about 3 minutes. At about 80% humidity in Mombasa it sticks for under 30 seconds or not at all. On humid days water vapour in the air lets the balloon's charge escape, so the balloon loses its charge quickly and the effect is weak. (c) A neutral electroscope has its leaf at 0 degrees. A charged object gives the cap, stem and leaf the same sign, so the leaf and stem repel and the leaf diverges. A rubbed balloon making the leaf rise shows that it is charged. With the electroscope negatively charged (about 30 degrees), a negative object raises the leaf above 30 degrees and a positive object lowers it below 30 degrees, so the sign can be found. A bigger angle shows more charge: 40 strokes gave about 50 degrees and 20 strokes gave about 30 degrees, because more electrons are transferred. Earthing with a finger returns the leaf to 0 degrees.
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Part 5: Applications and your complete Final Explanation

(a) Choose TWO applications from lightning conductors in Kericho and Kisumu, fuel tanker earthing at Nairobi petrol depots, photocopiers, and electrostatic precipitators at Athi River cement factories. For each one, explain which electrostatics idea from the unit it uses. (b) Now write a short, connected answer to the Driving Question: Where do electric charges come from? Why does a rubbed balloon attract or repel other objects? Include hair, the wall, paper, humidity, and how your thinking changed from your Lesson 1 model.	(a) Fuel tanker: a tanker trails an earthing chain. Moving fuel rubs against the tank and charge builds up by friction. Earthing lets electrons flow to the ground, so the charge cannot build up and cause a spark near fuel. Electrostatic precipitator: smoke particles are given a charge and are attracted to plates of the opposite charge. This is the rule that unlike charges attract. The particles are collected and not released into the air near Athi River. (Lightning conductor: point discharge and earthing. Photocopier: attraction between unlike charges.) (b) Electric charges are not created. They come from electrons that already exist in atoms. A neutral atom has equal protons and electrons. When I rub the balloon on dry hair, electrons transfer from the hair to the latex because latex holds electrons more strongly. The balloon gains about 1 nC, about 6×10^9 electrons, out of about 10^{23} atoms. The balloon becomes negative and the hair positive, and the total charge is conserved. The hair strands, all positive, repel each other and rise, and the unlike charges of the hair and balloon attract. Two rubbed balloons are both negative, so they repel. The negative balloon induces charge separation in the neutral wall and paper, with the near side positive. The attraction is stronger than the repulsion, so the paper jumps and the balloon sticks, and a neutral object is never repelled. Paper pieces that touch the balloon gain electrons, become negative and fall off. Latex is an insulator, so the balloon keeps its charge for about 3 minutes on a dry day. On humid days, such as in Mombasa at about 80%, water vapour lets the charge escape, so the effect is weak. In Lesson 1 my model only described what I saw. My final model shows electron transfer, charge separation and charge leaking away.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Origin of charge (electron transfer	Clearly explains that charges come from	Explains that electrons transfer from hair	Says that rubbing creates charge or does

and conservation)	electrons already in atoms, that electrons move from hair to latex and protons stay put, uses the triboelectric order and the 1 nC, 6×10^9 electrons, 10^{23} atoms figures, and states that charge is transferred, not created.	to balloon so the balloon is negative and hair positive, and mentions that charge is conserved, but gives few or no numbers or no reason from the triboelectric order.	not say what moves, or mixes up which object is positive and which is negative.
Attraction and repulsion, including neutral objects	Uses like-repel and unlike-attract with evidence from the rods and strips, and explains attraction of neutral paper and wall by induction (charge separation, nearer charge attracts more), with a labelled particle diagram and the point that neutral objects are never repelled.	Applies the rule to balloons and hair and mentions that neutral objects are attracted because of charge separation, but with limited detail or no diagram.	Only describes attraction and repulsion without giving a cause, or says that neutral objects are repelled or that positive charge moves.
Conductors, insulators, charging methods and distribution	Correctly links free versus tightly bound electrons to the conduction test and to the balloon keeping its charge, and compares contact (same sign) with induction and earthing (opposite sign). Explains charge on outer surfaces and at points with point discharge, using the electroscope and proof-plane readings.	Correctly states the difference between conductors and insulators and between contact and induction, and gives some of the readings, with some gaps in explaining distribution or earthing.	Confuses conductors with insulators or contact with induction, or gives no use of the readings.
Detection, humidity and applications	Explains the electroscope (same sign on leaf and stem, bigger angle means more charge, sign test, earthing), explains humidity using 35% and 80% and the 3 minutes against under 30 seconds figures, and links two applications to the correct idea.	Explains how the electroscope detects charge and mentions humidity and one application correctly, with few data values.	Does not explain how the electroscope works or the effect of humidity, or applications are not linked to electrostatics ideas.
Use of evidence, vocabulary and coherence	Uses exact data and lesson evidence, uses terms such as electric charge, electrostatic force, discharge, earthing and induction accurately, and gives a connected final answer that shows how the Lesson 1 model has changed.	Uses most evidence and terms correctly and gives a mostly connected answer.	Uses little evidence or incorrect terms such as static electricity as an explanation, and the answer is disconnected.

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